Keywords: video benchmark • omni-modal • temporal grounding • audio-visual understanding

OmniCompass: A Holistic Benchmark for Omni-Modal Video Grounding and Understanding

Paired-Setting Evaluation of 50 MLLMs Reveals Widespread Pattern-Matching and Persistent Grounding Failures in Both Vision-Language and Audio-Visual Tracks

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arXiv:2610.18829 • October 2026

Today's video benchmarks test language or vision in isolation, leaving the joint audio-visual grounding capabilities of omni-modal models unmeasured. *OmniCompass* introduces a paired evaluation protocol—each instance evaluated under text-only *and* text+evidence conditions—across two tracks: OmniCompass-VL (820 videos, 2,460 instances) and OmniCompass-AV (240 videos, 720 instances from narrative-rich, audio-dense content). Evaluation of 50 MLLMs finds that open-source models trail proprietary ones by 15–20 points on VL tasks and more on AV tasks; widespread paired-setting inconsistency (>30% of instances) exposes surface-level pattern matching in even the best-performing models.

AT A GLANCE

Problem: Video benchmarks test language or vision in isolation; no systematic measurement of joint audio-visual grounding exists.

Why it matters: Omni-modal models are deployed in video understanding; benchmarks must reveal whether they genuinely ground queries in evidence or pattern-match instruction formats.

Method: Two-track benchmark with paired evaluation (text-only/text+evidence); 8 VL task categories and 5 AV categories; temporal grounding, spatial grounding, and QA subtasks.

Result: 50 MLLMs tested; open-source trails proprietary by 15–20 pp on VL and more on AV; >30% paired-setting inconsistency in the majority of models.

Evidence: 820+240 videos, 2,460+720 instances; inter-annotator $\kappa > 0.8$.

The Big Picture

Omni-modal models can now process video, audio, and text simultaneously, yet their evaluation lags: benchmarks still test modalities in isolation. A model that answers a temporal grounding question correctly under text-only conditions may fail when the same question is accompanied by contradictory visual evidence—or vice versa. This mismatch is invisible to single-condition evaluation.

OmniCompass is built around a paired evaluation protocol that tests every instance twice: once with text alone and once with the relevant visual or audio evidence. A model that genuinely understands the target should respond consistently (and correctly) under both conditions. Inconsistency across conditions is a diagnostic signal: the model is pattern-matching the instruction format rather than grounding in the evidence.

Why Existing Approaches Fall Short

Standard VQA benchmarks test single-image comprehension; they do not cover video-level temporal or spatial grounding across a narrative.

Text-only LLM benchmarks ignore visual and audio modalities entirely; they cannot measure multimodal grounding by construction.

Existing video QA benchmarks (MVBench, VideoMME, etc.) focus on visual understanding without audio-dependent tasks or the paired-setting probe that would reveal instruction-format reliance.

Prior audio-visual benchmarks are small-scale and lack the structured temporal/spatial annotation and the paired consistency diagnostic that OmniCompass provides.

Core Mechanism: Paired Evaluation Design

Two evaluation tracks:

OmniCompass-VL (Vision-Language): 820 videos, 2,460 instances. Questions require visual evidence. Eight task categories: five perception (object recognition, action recognition, attribute identification, counting, OCR) and three reasoning (causal, commonsense, temporal reasoning). Three subtask types: temporal grounding (TG), spatial grounding (SG), and question answering (QA).

OmniCompass-AV (Audio-Visual): 240 videos, 720 instances from narrative-rich, audio-dense content. Questions require joint audio-visual evidence. Five task categories: three perception (speech recognition, sound identification, speaker localization) and two reasoning (audio-visual causal reasoning, event coordination). Same TG/SG/QA subtask structure.

Paired evaluation: Each instance is evaluated twice per model: text-only and text+evidence. Inconsistency

between conditions flags surface-level pattern matching.

Technical Formulation

For model $\mathcal{M}$, query q_i, visual/audio evidence v_i, and ground-truth answer a_i^*, define per-instance scores:

$$s_i^{\text{text}} = \mathbf{1}[\mathcal{M}(q_i) = a_i^*], \quad s_i^{\text{multi}} = \mathbf{1}[\mathcal{M}(q_i, v_i) = a_i^*]$$

The **consistency score** across N instances is:

$$C(\mathcal{M}) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\left[s_i^{\text{text}} = s_i^{\text{multi}}\right]$$

A high $C(\mathcal{M})$ with high $\sum_i s_i^{\text{multi}}$ indicates reliable grounding; a high $C(\mathcal{M})$ with low accuracy indicates consistent failure; a low $C(\mathcal{M})$ regardless of accuracy indicates format sensitivity.

Grounding precision is measured separately for temporal and spatial tasks. For a predicted interval $[\hat{t}_s, \hat{t}_e]$ and ground-truth $[t_s, t_e]$:

$$\text{T-IoU} = \frac{[\hat{t}_s, \hat{t}_e] \cap [t_s, t_e]}{[\hat{t}_s, \hat{t}_e] \cup [t_s, t_e]}$$

Spatial grounding uses standard bounding-box IoU.

Learning or Inference Procedure

OmniCompass is a benchmark; no model training is involved. Evaluation protocol:

1. Collect videos with expert-annotated temporal segments, spatial bounding boxes, and QA pairs (inter-annotator $\kappa > 0.8$).
2. Adversarially filter questions solvable from text alone.
3. Construct paired instances: text-only and text+evidence versions of every question.
4. Evaluate each model in zero-shot mode on all instances.
5. Compute per-track accuracy, T-IoU, spatial IoU, and consistency scores; stratify by task category and model family.

What the Guarantee Says

- Majority of evaluated models show **consistency scores below 70%**: over 30% of instances produce different correctness outcomes depending on whether evidence is provided.
- Grounding precision (temporal IoU, spatial IoU) remains **low across all model families**, including proprietary ones.
- Open-source models **trail proprietary by 15–20 pp** on the VL track; the gap widens on the AV track.
- Models that score well on VL tasks **do not uniformly transfer** to AV tasks, showing audio-visual reasoning requires more than visual competence.

Experimental Findings

- **Models (50 total):** Proprietary: GPT-4o, Claude Opus, Gemini models. Open-source: Qwen-VL, LLaVA variants, InternVL.
- **VL accuracy gap:** Proprietary average 15–20 pp above open-source on QA subtasks; smaller gap on perception, larger on reasoning.
- **AV accuracy gap:** The proprietary-to-open-source gap widens; several open-source models perform near chance on audio-visual causal reasoning.
- **Consistency:** Majority of models below 70%; temporal reasoning and causal reasoning show the lowest consistency.
- **Grounding:** T-IoU and spatial IoU are low across all families, confirming grounding—not just QA—is the bottleneck.

Ablations and Interpretation

Paired-gap analysis: Models with the smallest text-vs-multimodal accuracy gap tend to be the most accurate overall; genuine grounding correlates with benchmark performance.

Category breakdown: Temporal reasoning and causal reasoning show the lowest consistency scores, suggesting models rely heavily on surface-level linguistic cues for these harder categories.

Audio-visual transfer: Models that excel at VL tasks do not uniformly extend their advantage to AV tasks; audio-visual reasoning requires explicitly joint processing that visual-only training does not provide.

Adversarial filtering: Questions solvable from text alone inflate apparent model capability; after filtering, accuracy on the remaining paired set drops substantially, exposing true grounding rates.

Annotation quality: All task categories meet inter-annotator agreement $\kappa > 0.8$; the consistency-score diagnostic is robust to annotation noise because it compares a model against itself across conditions, not against potentially noisy labels.

Reference

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https://arxiv.org/abs/2610.18829